EI MegaProbe Language

a bioinformatics langauge

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Agenda

- Introduction
- Methods
- Current Results
- Conclusions
- Future Work

Introduction

- El MegaProbe Language: Statically-typed, ML-like language with bioinformatics primitives
- Help The Blueprint of Life project with a language that can handle large genome sequences

Methods

The language implementation follows the convention from Krishnamurthi [2].

- Lexing and parsing: parser combinators [1]
- Type Checking and Evaluating: Pattern Matching

Current Results

- An interpreter
- DNA and RNA sequences as primitives, and higher-order functions like transcribe DNA sequence and reverse complement.
- (minimal) Apache Arrow bindings for fast reading csv files and producing Apache Arrow Tables.

Conclusions

A fully functional interpreter has been built with most common programming constructs and primitives, with additional bioinformatics primitives and functions.

Future Work

- Add ADTs and pattern matching
- Necessary bindings to Apache Arrow
- The Plan Monad: How should we evaluate large dataframes? Laziness!
- Compile to GPU

Graham Hutton and Erik Meijer.
Monadic parser combinators, 2011.

Shriram Krishnamurthi.
Programming languages: Application and interpretation.
Brown University, 2017.